

Linear Algebra

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Source Code & Contributions

GitHub: <https://github.com/sihyeon8240/math>

Feel free to open issues or pull requests for corrections.

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Preface

Throughout this textbook, **logical formulas** are interpreted according to the following conventions:

- **Parentheses** have the highest precedence. For example,

$$(P \vee Q) \wedge R$$

is evaluated by first computing $P \vee Q$.

- The **logical connectives** are interpreted in the order

$$\neg > \wedge > \oplus > \vee > \implies, \iff > \iff$$

Hence

$$P \oplus Q \implies \neg R \vee S \wedge T \iff U \iff V$$

is understood as

$$((P \oplus Q) \implies ((\neg R) \vee (S \wedge T))) \iff (U \iff V)$$

- A **quantifier** binds the entire formula that follows it. Thus

$$\forall x, P(x) \wedge Q(x) \implies R(x)$$

means

$$\forall x ((P(x) \wedge Q(x)) \implies R(x))$$

- The symbol $\Downarrow$ is **not** a logical connective; it is used only to indicate the flow of a proof or an explanation. For example,

$$\begin{array}{c} P \\ \Downarrow \\ Q \end{array}$$

merely expresses that P , and therefore Q in the current discussion; it is not itself a proposition.

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1.3 Dimension of a Vector Space

1.4 Sums and Direct Sums

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12.1 Definitions

12.2 Separating Hyperplanes

12.3 Extreme Points and Supporting Hyperplanes

12.4 The Krein-Milman Theorem

Bibliography

- [1] Serge Lang, *Linear Algebra*, Springer Nature.